

Tutorial 01 Today

Please go to your tutorial room according to the last four digits of your student number shown below: (For example, if the last four digits of your student number is 8888, which is in the range 7887-9914, you are in Tutorial B4.)

Last four digits of student number	Tutorial section and room	TA name
0034-1827	Section B1, Room LA A720	Aaron Dwyer
1832-5267	Section B2, Room SA 406	Polina Vardashko
5318-7827	Section B3, Room NN 279	Biying Guo
7887-9914	Section B4, Room NN 280	Christopher Middlebrook

Test 1 on Oct. 01 during the tutorial

Test 1 (during the tutorial) on Oct. 01 will cover material from Notes 1-4 (chapter 1 of the textbook)

2.1 Functions

Example The temperature in Fahrenheit is related to the temperature in Celsius by the equation

$$F = \frac{9}{5}C + 32.$$

Recall that the graph of this linear equation is a straight line. For each input value of C the equation gives a **unique** output value of F . This is an example of a *function*.

We note that the variable C cannot take a value which is less than the absolute zero temperature -273.15 . We say that the domain of this function is $[-273.15, \infty)$.

2.1 Functions

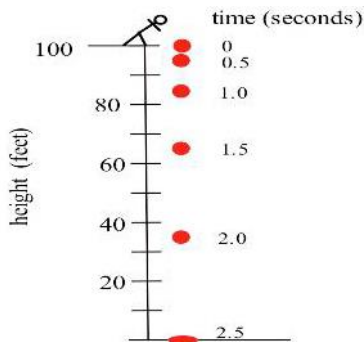
In general, a *function* f of a variable x is a rule that assigns to each input of x from a given set X a unique output y . We write $y = f(x)$ and call x the *independent variable* (*input* or *argument*) of the function, and y the *dependent variable* (*output*). The set X is called the *domain* of the function f and we write $Dom\{f\} = X$. We also call the set $\{f(x) : x \in X\}$ the *range* of f and denote it by $Range\{f\}$.

Example Each student at Carleton University is assigned a unique student number. This defines a function f , where $Dom\{f\}$ is the set of all Carleton students, and $Range\{f\}$ is the set of all student numbers.

2.1 Functions

Example The height of a falling tomato is given by the function $h = f(t) = 100 - 16t^2$, where t is the time (in second) after the tomato is released and h is the height (in feet) above the ground.

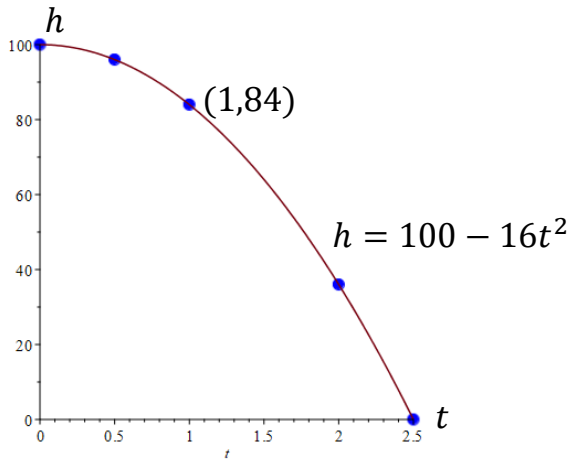
The domain of the function f is $Dom\{f\} = [0, 2.5]$,
The range of this function is $Range\{f\} = [0, 100]$.



time (s)	height (ft)
0	100
0.5	96
1.0	84
1.5	64
2.0	36
2.5	0

Graph of a Quadratic Function

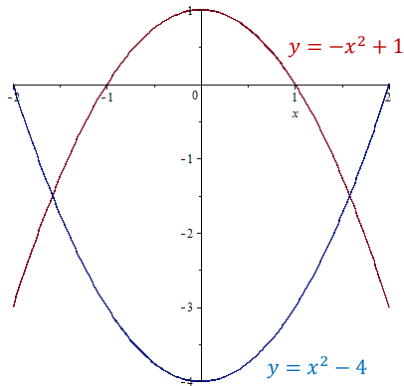
t	0	0.5	1.0	1.5	2.0	2.5
h	100	96	84	64	36	0



Quadratic Functions

For constants a, b, c with $a \neq 0$, $f(x) = ax^2 + bx + c$ is called a *quadratic* function, its graph is a parabola.

When $a > 0$, the parabola opens upward, and the lowest point of the parabola is called the *vertex* of the parabola; when $a < 0$ the parabola opens downward, and the highest point of the parabola is the *vertex* of the parabola.



Quadratic Functions

How to find the vertex of the parabola given by the quadratic function $f(x) = ax^2 + bx + c$?

Completing square:

$$\begin{aligned}f(x) &= ax^2 + bx + c \\&= a \left(x^2 + \frac{b}{a}x \right) + c \\&= a \left(\left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a^2} \right) + c \\&= a \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a}.\end{aligned}$$

This is called the *vertex form* of the quadratic function, and the vertex is $\left(-\frac{b}{2a}, \frac{4ac - b^2}{4a} \right)$.

Quadratic Functions

Example Plot the graph of $y = x^2 - 5x + 6$. Find its vertex and x -intercepts (if there is any).

Solution We first rewrite the quadratic function in the vertex form (with $a = 1, b = -5, c = -6$):

$$\begin{aligned}y &= x^2 - 5x + 6 = \left(x - \frac{5}{2}\right)^2 + \frac{4 \times 6 - (-5)^2}{4 \times 1} \\&= \left(x - \frac{5}{2}\right)^2 - \frac{1}{4}.\end{aligned}$$

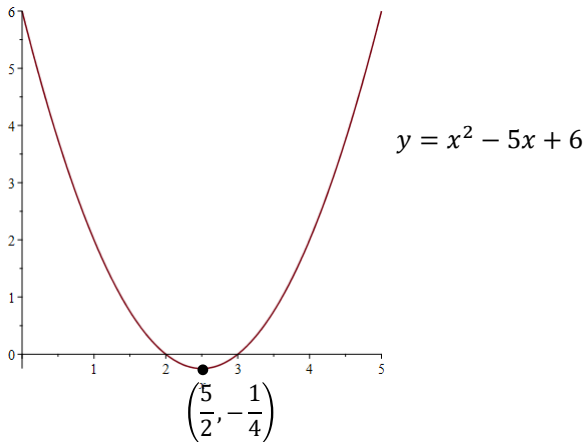
Hence the vertex is $(5/2, -1/4)$. The x -intercepts are found by setting $y = 0$. That is,

$$\left(x - \frac{5}{2}\right)^2 = \frac{1}{4}.$$

Hence $x - \frac{5}{2} = \pm \frac{1}{2}$, which gives $x = 3$ or $x = 2$.

Quadratic Functions

Remark: The x -intercepts can also be found by factoring $y = x^2 - 5x + 6 = (x - 2)(x - 3)$. The graph is shown below.



Quadratic Functions

Example Plot the graph of $y = 2x^2 - 2x - 12$. Find its vertex, y -intercept and x -intercepts.

Solution We first rewrite the quadratic function in the vertex form (with $a = 2$, $b = -2$, $c = -12$):

$$\begin{aligned} 2x^2 - 2x - 12 &= 2 \left(x - \frac{1}{2} \right)^2 + \frac{4 \times 2 \times (-12) - (-2)^2}{4 \times 2} \\ &= 2 \left(x - \frac{1}{2} \right)^2 - \frac{25}{2}. \end{aligned}$$

Hence the vertex is $(1/2, -25/2)$.

Quadratic Functions

The y -intercept is found by setting $x = 0$, which gives $y = -12$.

The x -intercepts are found by setting $y = 0$. That is,

$$2 \left(x - \frac{1}{2} \right)^2 = \frac{25}{2} \implies \left(x - \frac{1}{2} \right)^2 = \frac{25}{4}.$$

Hence

$$x - \frac{1}{2} = \pm \frac{5}{2} \implies x = 3, -2.$$

The graph of $y = 2x^2 - 2x - 12$

